

Worksheet W6: The Odometer Trick

"Adding up tiny slices to find total distance."

Chapter 6 · 10 Problems · Name: _____ Date: _____

The Integral = Total Accumulation = Adding up all the tiny changes

Problem 1

Question: What is the **integral** in calculus?

💡 *Hint: Think about an odometer — it adds up all the tiny pieces.*

Problem 2

A car drives at 50 mph for 2 hours and then 70 mph for 1 hour. What is the total distance?

💡 *Hint: Distance = Speed $\times$ Time for each part, then add.*

Problem 3

Question: What does the **area under a speed curve** represent?

💡 *Hint: Speed is height, time is width. Area = height $\times$ width = ?*

Problem 4

A car's speed over time is shown below. Estimate total distance using 1-hour intervals.

Time (hours)	0 to 1	1 to 2	2 to 3	3 to 4
Speed (mph)	50	60	70	40

💡 Hint: $\text{Distance} = \text{Speed} \times \text{Time}$ for each interval, then add.

Problem 5

Question: What is the **integral symbol** and what does it mean?

💡 Hint: It looks like an elongated 'S'. It stands for "sum."

Problem 6

Question: If you divide the area under a curve into **4 rectangles** instead of 1, does the estimate get better or worse?

💡 Hint: More rectangles = ?

Problem 7

Estimate the area under $y = x$ from $x = 0$ to $x = 4$ using **4 rectangles** (width = 1). Use the left-hand sum.

Interval	0 to 1	1 to 2	2 to 3	3 to 4
Height ($y = x$)	0	1	2	3
Area (height $\times$ 1)	0	1	2	3

💡 Hint: Add the areas: $0 + 1 + 2 + 3 = ?$

Problem 8

Question: In the integral notation $\int f(x) dx$, what does dx represent?

💡 *Hint: It's a tiny piece of x — the width of a rectangle.*

Problem 9

A car accelerates smoothly from 0 to 60 mph over 3 hours. The speed curve is a straight line from (0,0) to (3,60). What is the total distance?

💡 *Hint: The area under the line is a triangle. $\text{Area} = (\text{base} \times \text{height}) \div 2$.*

Problem 10

Question: Complete this sentence: "The derivative and the integral are _____."

💡 *Hint: Think about addition and subtraction — they undo each other.*